Divita Saraogi

Data Scientist | AI & Machine Learning Engineer

📧 divitadsaraogi@gmail.com | 📱 +91-8847759169 | Bengaluru, India
LinkedIn: linkedin.com/in/divita-saraogi | Github: github.com/DivitaDSaraogi

PROFESSIONAL SUMMARY

Machine Learning Engineer and Data Scientist with a Ph.D. from IIT Bombay and 2+ years of industry experience building scalable AI solutions for industrial and scientific datasets. Experienced in production ML pipelines, deep learning, feature engineering, distributed data processing, and statistical modelling using Python, PyTorch, PySpark and AWS. Currently developing AI-driven spectral intelligence systems at Picarro for large-scale chemical compound identification.

TECHNICAL SKILLS

Programming Languages	Python • SQL • R • C++ • C
Machine Learning & AI:	PyTorch • TensorFlow • Scikit-learn • Deep Learning • Machine Learning • Feature Engineering • Model Optimisation • Classification • Regression • Clustering • PCA • Dimensionality Reduction • Model Evaluation • Hyperparameter Optimisation
Data Engineering:	PySpark • Pandas • NumPy • SciPy • Dask • PyArrow • ETL Pipelines • Data Processing • HDF5 • Parquet • Distributed Computing
Visualization:	Power BI • Matplotlib • Plotly • Seaborn
Cloud & Dev Tools:	AWS • Docker • Git • GitHub • Jupyter Notebook • Linux • PyCharm • Jira
Domain Expertise:	Spectral Analysis • Signal Processing • Scientific Computing • High-Dimensional Data Analysis • Feature Extraction •Machine Learning Systems • Time-Series Analysis
Statistics:	Bayesian Statistics • Statistical Modelling • Hypothesis Testing • Experimental Design • Probability • Regression Analysis

PROFESSIONAL EXPERIENCE

Data Scientist – AI Team | Picarro Technologies India Pvt. Ltd.

Jun 2025 – Present | Bengaluru, India

- Developed AI-driven spectral intelligence pipelines for compound identification and quantification using **Python, PyTorch, and AWS**.
- Built ML and deep learning models for identifying **600+ chemical compounds** from high-dimensional spectroscopy data.

- Processed and analysed **100M+ spectral observations**, developing scalable feature engineering and data-processing pipelines.
- Collaborated with cross-functional teams to develop production-ready machine learning solutions.

Data Scientist | Tosscall Services India Pvt. Ltd.

Feb 2024 – Feb 2025 | Bhilai, India

- Developed machine learning models for customer return prediction and business analytics.
- Performed EDA, feature engineering, and statistical modelling on large business datasets.
- Built **Power BI** dashboards and optimised SQL workflows to deliver actionable business insights.

PhD Research Scholar | Indian Institute of Technology Bombay

Jan 2018 – Jan 2024 | Mumbai, India

- Developed automated localisation pipelines for **Gamma-Ray Bursts** using AstroSat satellite data.
- Built scalable scientific data-processing pipelines using Python and statistical modelling techniques.
- Published **12 peer-reviewed research papers** and presented research at international conferences.

EDUCATION

Ph.D. in Physics, Indian Institute of Technology Bombay (IIT Bombay), Mumbai |2018 - 2024
M.Sc. in Physics, Department of Physics, Sant Gadge Baba Amravati University , Amravati| 2012 - 2014
B.Sc. in Physics,Chemistry, Mathematics from Lady Amritbai Daga (LAD) College, Nagpur | 2009 - 2012

PUBLICATIONS

- **12 peer-reviewed international publications.**
- Google Scholar profile available upon request.

AWARDS & ACHIEVEMENTS

- ASI Best Poster Award (2024) for Gamma-Ray Burst localization research.
- Invited Speaker – Machine Learning & AI, MVJ College of Engineering, Bengaluru.

CERTIFICATIONS

Microsoft SQL • Python for Data Science • Power BI • Statistics & Mathematics for Data Science • Advanced Excel